K. J. SOMAIYA INSTITUTE OF TECHNOLOGY

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Comparative Time-Series Forecasting of Food Inflation using El Niño Indices: ARIMA vs. Prophet vs. LSTM

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Abstract

This project investigates whether El Niño climate information improves forecasts of Indian food inflation. Monthly food-inflation and Oceanic Niño Index (ONI) data from 2012–2025 were cleaned, aligned and used to develop four time-series models: ARIMAX, Prophet, LSTM with ONI, and an inflation-only LSTM baseline. All models were evaluated on the same January 2024–December 2025 holdout period using MAE, RMSE, sMAPE and MASE. The baseline LSTM achieved the lowest forecasting error, and the results are presented through a FastAPI-powered interactive web dashboard.

Background & Problem Statement

Food inflation directly affects household purchasing power, food planning and short-term policy responses. El Niño can influence rainfall and agricultural conditions, making climate indicators a plausible input for inflation forecasting. However, weather-related signals do not necessarily improve predictive accuracy once historical inflation patterns are considered. This study addresses that gap by comparing ONI-aware models with a history-only baseline under the same unseen test window and by making the outputs accessible through a web dashboard.

Objectives

- Prepare and align monthly Indian food-inflation data with the Oceanic Niño Index.
- Implement ARIMAX, Prophet, LSTM + ONI, and an inflation-only LSTM baseline.
- Evaluate all four models over January 2024–December 2025 using four error metrics.
- Compare the incremental forecasting value of ONI against the historical-only baseline.
- Present model predictions, performance scores and insights through a web dashboard.

Dataset & Evaluation

Data: 168 monthly observations (January 2012–December 2025) with processed variables including date, food inflation (%), ONI, and month.
Training period: January 2012–December 2023 (144 months).
Test period: January 2024–December 2025 (24 months).
Metrics: MAE, RMSE, sMAPE, MASE (lower is better).
Fairness rule: all final models were compared on the same chronological holdout window.

Research Question

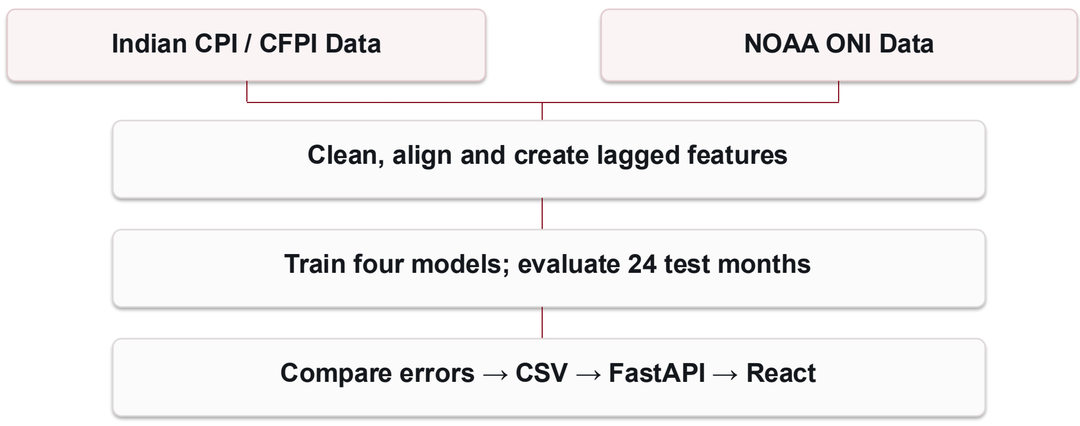
Does adding ONI to an LSTM improve out-of-sample food-inflation prediction compared with using inflation history alone?

Observed benchmark result: No. In this study, the inflation-only LSTM remained the best-performing model.

Study Limitations

The result is specific to the available dataset and 24-month test period. Climate effects may be delayed or mediated by many other economic factors. Forecast horizons and information used at test time also differ between model families. The work evaluates historical holdout performance only; prospective future-month validation and live model retraining were outside the present scope.

Methodology



Technology Stack

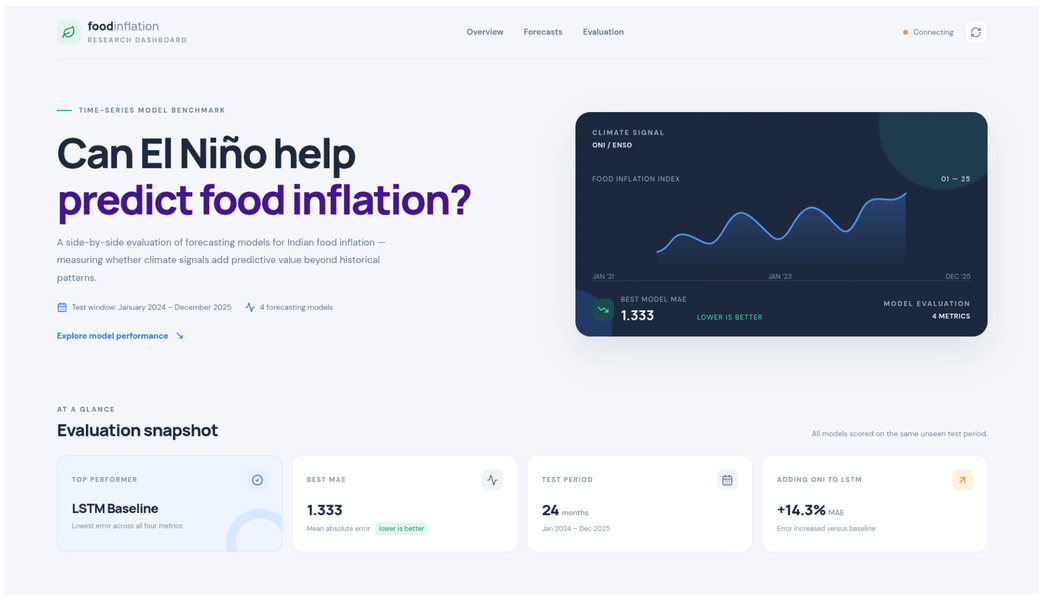
Models: ARIMAX (SARIMAX), Prophet, LSTM + ONI, LSTM Baseline
ML tools: Pandas, NumPy, Scikit-learn, Statsmodels, TensorFlow/Keras
Application: FastAPI, React, Vite, Recharts, Express
Deployment: Render API and Vercel web dashboard
Outputs: processed CSVs, REST endpoints, interactive evaluation dashboard

Results and Discussion

Historical test period: January 2024 – December 2025

Model	MAE	RMSE	sMAPE	MASE	Rank
LSTM Baseline	**1.3335**	**1.6460**	**43.89%**	**0.9910**	**1**
LSTM + ONI	1.5245	1.9422	49.64%	1.1330	2
ARIMAX	4.6550	6.3055	85.44%	3.4594	3
Prophet	5.0169	6.6850	87.40%	3.7284	4

The inflation-only LSTM ranks first on all four metrics. Adding ONI increased LSTM MAE from 1.3335 to 1.5245, so the climate signal did not improve final out-of-sample accuracy in this benchmark.



Web dashboard interface preview (based on the project frontend design)

Conclusion

The LSTM baseline provided the most accurate predictions in the selected test period, while ONI did not improve the final performance of the LSTM benchmark. The developed dashboard makes this comparison accessible and interpretable through tables, graphs and forecast views. These findings do not establish that El Niño has no influence on food prices; instead, they show that ONI alone did not outperform the history-only baseline in the present setup.

Acknowledgement

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